The difficulty of diagnostics

Tests as evidence rather than Truth

The signal gets swamped: Bayes rule and prevalence

Let us continue thinking about diagnostic testing. Consider antibody-based tests for Covid-19 infections. There was surprisingly little publicly available information published about diagnostic sensitivity[1]. In part this was because it was not clear how to determine who is *truly* infected and who is *truly* negative[2] and in part it is because the amount of virus (and antibodies) changes throughout an infection, varies among different classes of people, and so on.

[1] As opposed to analytic sensitivity, which measures how little of a pathogen can be detected.

[2] Usually one compares their test to a "gold standard," which is assumed to be correct, but in this case with a novel disease how can one be sure? This is never an easy problem, but it's even harder in the middle of a fast-moving crisis!

But let's simplify things and instead consider a real-time reverse-transcriptase PCR reactions, the basis of many diagnostic tests, include that for Covid-19. These are very good tests, so let's imagine it has a sensitivity of 0.95 (95%) and a specificity of 0.97 (97%). Not bad, right? (Most tests, especially those based on detecting viral antigens or antibodies, are almost certainly worse, especially on the sensitivity side of things, but let's just pretend.)

But here comes the wrinkle. Let's assume that 1% of the population as a whole is infected. If we were to randomly[3] test 10,000 people we would expect 100 people to be really, truly infected and of those, 95 would be correctly identified[4], leaving 5 false negatives[5] (Figure 1). I guess we could live with that, although if we were thinking of Ebola virus infections that might be worrying!

[3] This is important in that if we just test sick people we would tend to have more infected people than the population average of 1%.

[4] 100×0.95

[5] $100 \times (1 - 0.95)$

Now what about the 9,900 uninfected people? The vast majority, 9,603, would be correctly told they are not infected[6], but 3% of them, or 297 uninfected people would be told they *are* infected[7]. Or put another way, only 95/(95+297) = 24% of the people who receive a positive test are actually infected! Our pretty darn good test is telling a bunch of people they are infected when they are in fact not! Indeed, in this scenario if you get a positive test result, the odds are about 3:1 that you are *not* infected!

[6] 9900×0.97

[7] $9900 \times (1 - 0.97)$

But wait, it can be even worse. In early stages of the epidemic when perhaps only 0.1% of the population was infected, nine or ten[8] of the truly infected individuals would be detected, but roughly 300 of the 9,990 *uninfected* people would get false positives[9], so only 10/(10+300) = ~3% of the positive tests would indicate true infections.

[8] $10 \times 0.95 = 9.5$. Let's round up to 10.

[9] $9990 \times (1-0.97) = 299.7$, rounding up to 300.

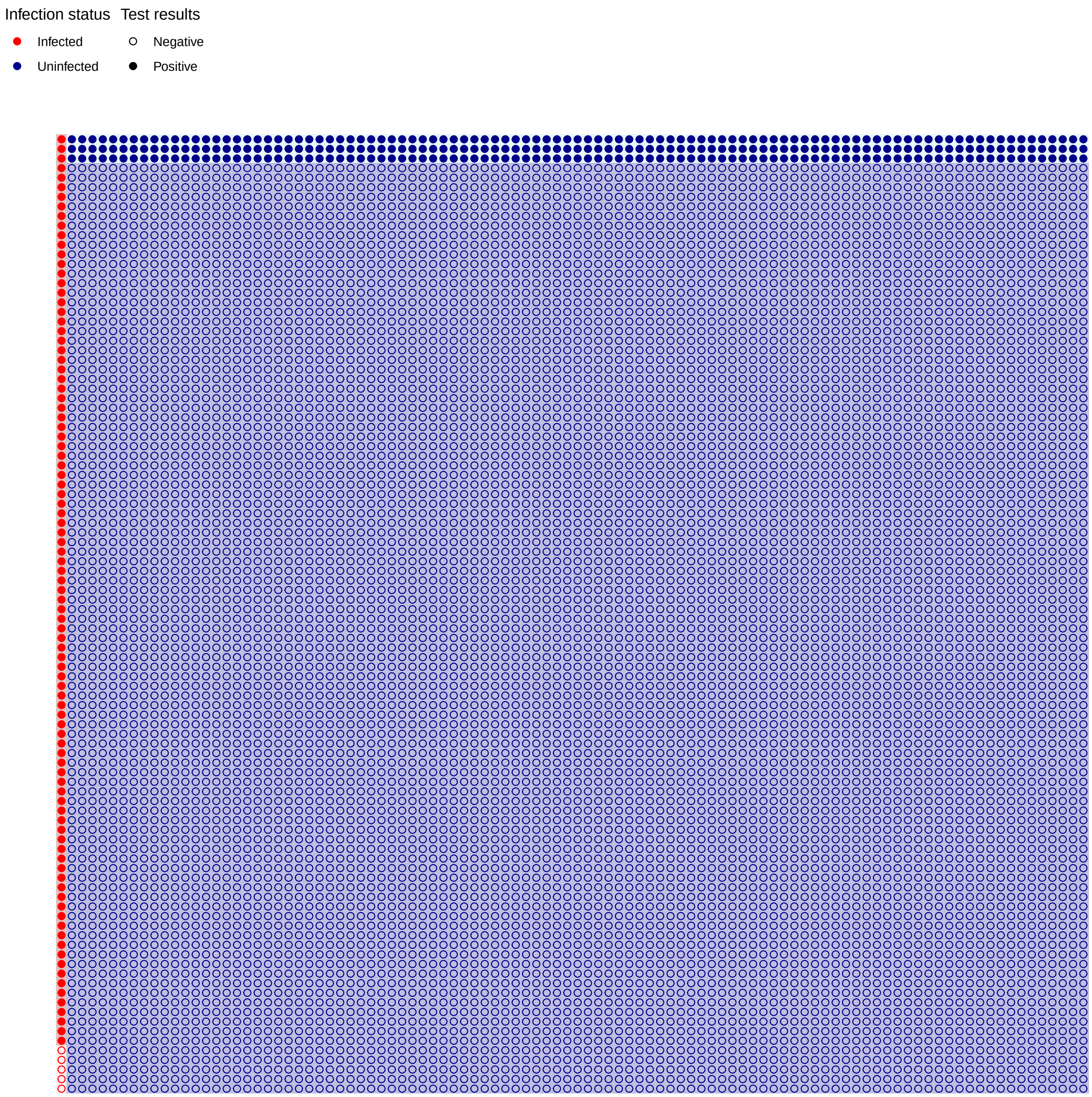

Figure 1: Ten thousand individuals (dots), 1% (100) of which are infected (red) and 99% (9,900) that are negative. With a sensitivity of 95% we should get 95 positive tests from the truly infected, but because the specificity is 97% (meaning 3% false positives) we will get 297 (9900×0.03) false positives from the uninfected individuals (blue).

This is often presented as "The medical paradox," though it isn't really a paradox. It is also used as an example of Bayes theorem. You might come across it as[10]:

[10] This is included for completeness, not because seeing it this way is especially edifying unless you really love conditional probabilities, in which case, it's your lucky day!

$$\Pr(I^+|T^+) = \frac{\text{Sensitivity} \times \text{Prevalence}}{\Pr(\text{True positive}) + \Pr(\text{False positive})} \tag{1}$$
$$= \frac{\text{Sensitivity} \times \text{Prevalence}}{\text{Sensitivity} \times \text{Prevalence} + (1 - \text{Specificity}) \times (1 - \text{Prevalence})} \tag{2}$$

which we can read as: "The probability a person is infected, given that they test positive, is equal to the probability they test positive given that they are infected times the probability they really are infected, divided by the probability they test positive." Does that help? No? Didn't think so. We'll move on, but just for clarity, let's plug in the numbers from our example with 1% prevalence, a sensitivity of 0.95, and a specificity of 0.97:

$$\Pr(I^+|T^+) = \frac{0.95 \times 0.01}{0.95 \times 0.01 + (1 - 0.97) \times (1 - 0.01)} \tag{3}$$
$$= 0.242, \tag{4}$$

which matches our previous calculation.

Tests provide evidence, not absolutes

Sometimes people look at these examples and come to the conclusion that anything other than perfect diagnostic tests are worthless. But that's actually missing the point. Diagnostic tests are not magic wands bringing Truth, but rather they bring *evidence* to the table.

Let's walk through this example, but now paying attention to the *odds* a person is infected before and after the test[11].

[11] This way of presenting the medical paradox comes from a wonderful video by 3Blue1Brown. I think it makes the logic much clearer!

First, let's consider the odds a random person from our population of 10,000 is infected, prior to testing them. Not knowing anything else about them (e.g., whether they are high or low risk, young or old, etc.) we would probably guess their probability of being infected is 1%, the prevalence of infection in the population. This means the *odds*[12] of infection are $0.01/0.99 = 0.01010101$, or really, really small.

[12] Odds of infection $= \frac{\Pr(\text{infected})}{\Pr(\text{not infected})}$.

Now let's think about our test results, specifically, where they might have come from. A positive test might be a true positive or a false positive. The former, true positives, happen with a probability of 0.95, the sensitivity of the test, while the latter, the false positives, occurs with a probability of 0.03 (=1 - the specificity of the test). We can express the relative probability of these two hypotheses—the probability of getting a positive test from an infected person vs. the probability of getting a positive test from an uninfected person—as a ratio[13]. In this case, this ratio is $0.95/0.03 = 31.667$. That is, positive tests are *much* more likely to be true positives than false positives.

[13] Often called a Bayes factor, but let's not get distracted.

We can do the same thing with negative tests. Here probability of getting a negative test from an infected person is 1 - sensitivity, or $1 - 0.95 = 0.05$ and the probability of getting a negative test from an uninfected person is the test's specificity, here 0.97. Thus, the ratio is $0.05/0.97 = 0.0515$. Negative tests are *much* less likely to be false negatives than true negatives.

The cool part is that we can "update" our initial odds that a random person from our population is infected given these test results. We simply multiply the odds they were infected *prior* to the test by the ratio we calculated of getting the positive (or negative) test. So if our random person gets a positive test result, the odds they are infected becomes $\frac{0.01}{0.99} \times \frac{0.95}{0.03} = 0.0101 \times 31.667 = 0.320$. That is, their odds of being infected increased over thirty-fold after testing positive. Moreover, these odds of 0.320 corresponds to a probability of ~24%, which is the same as we calculated from the numbers in Figure 1, above.

If instead they had a negative test we would update their odds of infection as $\frac{0.01}{0.99} \times \frac{0.05}{0.93} = 0.0101 \times 0.0515 = 0.00054$; their odds *decreased* almost 20-fold! The probability of being infected went from 1% down to ~0.05%.

The point is that we are getting a lot of evidence from these tests. Better tests will have larger ratios, meaning they provide more evidence. But evidence, of course, is not certainty. Think about it this way: imagine you took a test for Ebola virus infection, a really, really good, accurate one. Now imagine you got a positive test. Would you really think that you were infected with Ebola virus? Really?! My guess is that you probably would not worry about it because your prior (before the test) probability of being infected with Ebola virus would be vanishingly small. But this test result might want you to check in with your doctor, right? Because it did update your beliefs a bit. This is probably a much more reasonable way to think about diagnostic tests.

One last note on this perspective: we or our (future) patients are usually not some random person. We often know a bit more about people, such as their age or occupation or where they live or other risk factors. We can (and do[14]) update our priors based on these factors. If, during the pandemic, you were masking up, staying home, and minimizing your risk, you might reasonably assume that your prior (before the diagnostic test) odds of infection were lower than our 0.01/0.99 calculation based on prevalence in the population, whereas if you were going to parties or had housemates who were infected, your prior odds of infection might be much higher. The *test* results still provide the same *evidence*, but your starting point would be different and so the odds you were or were not infected would be different.

[14] Though often informally.

This framing also helps explain how different people can look at the same exact diagnostic test (or evidence in a court case or application for benefits or…) and come to radically different conclusions. If we start with different prior beliefs, we'll end up in different places even if we agree on what the test result tells us!

Cool, huh? I know it was a long journey to get here, but I hope it was worth the trek through these prose and ideas!

What to do with diagnostic tests?

I hope I've convinced you that the seemingly simple task of sorting individuals into bins—infected or uninfected, or any other distinction—is actually much more difficult than it first seems. There are always gray areas and trade-offs lurking. Sure, we might improve our test sensitivity, but we do so at the cost of specificity. And depending on the context, a positive test might not translate into a high probability that you are actually infected. So, what's a person to do?

First, it is worth thinking about what we are trying to accomplish with a test. Tests are used for all sorts of things and pretending they are used in only one way ignores all of the trade-offs and potential biases we just learned. If we are thinking about you in a doctor's office getting back test results (or conversely, you, the doctor, giving someone the news) it is worth thinking about how well the test performs, but also your *prior* odds of having whatever you are testing for in the first place. You might also look for some confirmation from other independent tests or context. In short, tests provide useful *evidence*, but they are not magic wands!

Alternatively, if we are interested in trying to estimate the prevalence of infection in the population, we can correct our estimates with known diagnostic test sensitivity and specificity values. We may not know whether *this* person or *that one* is infected, but we could have a pretty good idea that, say, 20% of the population was infected, on average[15].

[15] This is part of the rationale for using fast, cheap, but crappy antibody-based tests for influenza or COVID-19. Yes, they suck, but they can still give us a window on what's happening and, importantly, we get the answers back quickly!

Second, it is worth noting that we can be clever about our testing regime. For instance, we might employ a two-step approach for widespread testing. We could employ a test with high sensitivity, but low specificity, for instance an antibody test tuned for maximum sensitivity. This would ensure that few truly infected people test negative, but would yield a *lot* of false positives. Those positives would then all be re-tested with a separate test with much higher specificity (e.g., a nucleic acid-based test) to sort out which of that big heap of "positives" from the first test were actually likely true positives. Most HIV screening is done this way.

Third, it pays to think about the costs of false negatives *and* false positives[16]. What are the costs of false positives during a pandemic? Perhaps many, many people being quarantined for no good reason, not to mention all of the stress and worry. What are the costs of missing a COVID-19 infection? A person might ignore their other symptoms because they thought they were clear, perhaps even die. At a larger scale, the infected person who thinks they are COVID-free might go on to infect (many) others and so the infection spreads.

[16] My own opinion is that many conflicts boil down to some people focusing on one set of costs—such as missing a tumor or someone who honestly needs assistance—while others focus only on the costs of false positives—unnecessary health care interventions or welfare cheats.

This highlights one other issue that crops up frequently when it comes to diagnostic testing: the right or optimal answer at an individual level is not necessarily the right or optimal answer for the population. A doctor might see the ghost of a tumor in a mammogram and recommend a biopsy if not a lumpectomy. Yes, it might have been a false positive, but the cost—an unnecessary procedure—seems small compared to letting a real tumor grow and, potentially, become much worse. They *care* for their patient after all! An epidemiologist, who focuses on the whole population, might instead note that because tumors are fairly rare and mammograms imperfect, there are lots and lots of false positives, the costs of which collectively outweigh the benefits of catching a few tumors early. They *care* about maximizing the health of the population as a whole! Both are "right," given their perspective, but they might strongly disagree about what is best. You could extend the same thinking to the TSA's security at airports or the Innocence Project's efforts to find and exonerate the wrongly convicted. Or even to all of those amazing people at offices across the nation trying to sort out who, among the many applicants, most need public assistance.